

&j OMNI-GRID

The Cognitive Energy Infrastructure

O Sistema Operacional da Energia Elétrica

13 REST + WS Endpoints | 10 ESP32 Modules | 30+ Android Screens | ~302 Tests | 8 CI/CD Jobs

JARBAS-OMNI-SOVEREIGN-INFINITY v5.1

Ø=Ý% O APAGÃO SILENCIOSO

Ø>Ý† Duck Curve

Solar oversupply at noon, demand peak at sunset. Grid can't adapt.

&¡ Frequency Instability

Renewables lack physical inertia. Blackout risk in milliseconds.

Ø=Ü° CAPEX Bottleneck

Utilities need trillions in new substations. Or they use Omni-Grid.

\$2T annual global electricity spend — yet the grid can't handle EVs + solar

Ø>Ýà A SOLUÇÃO: OMNI-GRID

Ø=Üá PDE Engine

Holt-Winters + Kalman
Monte Carlo 10k scenarios
CCEE/ONS Live Data

Ø=Üæ Omni-Box Hybrid

Android + ESP32-S3
5-transport fallback chain
Shadow Autonomous Mode

Ø=ÜÊ Dashboard

3 profiles (Ind/Res/Solar)
WebSocket Real-time
PWA + Multi-idioma

Não somos concorrentes — somos Redutores de CAPEX para as distribuidoras

Ø>ÝÊ PREDICTIVE DISPATCH ENGINE

Ø=ÜÈ Forecast Engine ' IMPLEMENTED

- Holt-Winters Triple Exponential Smoothing
- Kalman Filter for real-time correction
- Seasonal Naive Fallback (MAE < 5%)
- Ø=Ý Transformers + LSTM (Roadmap)

Ø<ß² Stochastic Optimizer ' IMPLEMENTED

- Monte Carlo with 10,000 price paths
- PLD price band simulation per submercado
- SoH degradation cost: 0.001%/h of dispatch

Loss: ! = ; u &öf—B !"²·BatteryDegradation " ;;tw idInstability

Ø=Üá Live Data Collectors ' IMPLEMENTED

CCEE — PLD 2021-2026
JSON API !" pricePerKwh (R\$/kWh)

ONS — Curva de Carga
CSV !" LoadRecord per submercado

ANEEL SIGA — Geração
CSV !" SigaGenerationAsset[]

Ø=Üæ OMNI-BOX: EDGE GATEWAY

Ø>Ý Android App (47 tasks)

Kiosk/DPC — Device Owner, PIN SHA-256

ConnectionManager — 5-transport fallback

TelemetryDatabase — SQLite store-and-forward

MeterReader — CameraX + OCR

OmniBoxFcmService — FCM push (4 types)

PowerManager — 4 battery profiles

TimeFallback — NTP!'GPS!'Cellular!'Uptime

OmniBoxViewModel — StateFlow UI

&™p ESP32-S3 Co-Proc (10 modules)

Modbus RTU/TCP — Port 502, 8 regs

CAN Bus 2.0 — JBD/Daly/JK @ 250kbps

BLE GATT — UUID 4f4d4e49..., 3 chars

WiFi AP — Omni-Box-AP + REST APIs

IEC 61850-9-2LE — SV multicast port 6000

Safety IEEE 1547 — Trip <100ms

Shadow Mode — RTC buffer 60 entries

OTA — ArduinoOTA + HTTP binary

Fallback: CDC !' BLE !' WiFi !' USB !' Offline | TelemetryFrame 34B @ 115200 baud | HW Watchdog GPIO47 @ 1s

Ø>Ý€ RUST CORE + CYBER-FORTRESS

Rust Crate (1,066 lines) ' 1

- lib.rs — Core state machine
- jni_bridge.rs — 13 JNI functions
- shadow.rs — RTC shadow engine
- safety.rs — IEEE 1547
- 28 inline + 23 integration tests

Ø=Þáp Security Layers

- ' mTLS X.509 certs per device
- ' SHA-256 PIN verification
- ' JWT HMAC-SHA256 with RBAC
- ' Zero Trust Architecture
- Ø=Ý Blockchain audit trail
- Ø=Ý Rate limiting + DoS

Ø=Üd RBAC: admin · operator · viewer · device · api_client

Ø=Ý API GATEWAY — FASTIFY :3000

Method	Route	Auth
GET	/health	No auth
POST	/auth/login	No auth
GET	/auth/me	JWT
WS	/ws?token=	JWT
POST	/api/v1/pde/forecast	JWT
POST	/api/v1/pde/optimize	JWT
POST	/api/v1/pde/dispatch/execute	No hook
GET	/api/v1/pde/dispatch/history	JWT
GET	/api/v1/market/prices	JWT
GET	/api/v1/market/submercados	JWT
GET	/api/v1/market/regulatory	JWT
POST	/api/v1/edge/telemetry	JWT
GET	/api/v1/telemetry/latest	JWT

Seed users: admin / operator / viewer / device-sim-001

Ø=Ü° UNIT ECONOMICS

Industrial 100kWh

Arbitrage (80kWh × R\$0.60)	R\$ 1,440/mo
Peak Shaving (30%)	R\$ 1,500/mo
Ancillary Services	R\$ 500/mo
Total MRR	R\$ 3,440/mo
Payback	3.6 years
LTV (12yr)	R\$ 240k

Utility Scale 1MWh

Revenue	US\$ 180-250k/yr
O&M Cost	US\$ 15k/yr
CAC	US\$ 30k
LTV (12yr)	US\$ 2.5M
LTV/CAC	> 80x
Gross Margin	85%+

Break-even M14 | MRR M12: R\$ 340k | 100 units | 5x growth M9!’M12

Ø<ÝçØ<Ý÷ BRAZILIAN ENERGY MARKET

4 Submercados PLD

SE/CO	SP,RJ,MG,ES,DF,GO,MT,MS
Sul	PR,SC,RS
Nordeste	BA,PE,CE,RN,PB,AL,SE,PI,MA
Norte	PA,AM,RO,AC,RR,AP,TO

11 Utilities + 5 Traders

~60%	Enel SP · CEMIG · CPFL · Light · EDP · Copel · Celesc · Coelba · Neoenergia PE · Equatorial PA · Energisa MT
~15%	
~15%	Traders: Tradener · Comerc · Ecom Energia · Safira · Delta Energia
~10%	

PLD: Piso R\$ 69,07 | Teto R\$ 599,31/MWh

Bandeira: Verde (seca) | Amarela +R\$18,85 | Vermelha +R\$44,63 / +R\$78,77

&™p IMPLEMENTATION STATUS + ROADMAP



~302 tests | All 4 sprints complete | 47 edge tasks done

Ø<β⁻ EXIT STRATEGY

Ø=ρâρ Big Oil

Shell, BP, Equinor — Hydrocarbon to electron transition

& ρ Big Tech

Amazon, Google, Microsoft — Data center load management

Ø=ρ— EV Giants

Tesla, BYD, VW — Software verticalization on batteries

Ø=ÛÈ IPO NASDAQ

First data-driven utility — valued at > US\$ 1.5 billion

"O bilhão não é o fim — é o combustível."

